Rajeev Jain

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Principal Research Software Engineer, Argonne National Laboratory

Scientific software for supercomputers, and tooling that lets AI agents use it. At Argonne since 2009.

Professional Experience

Argonne National Laboratory **2009 – Present**
Principal Specialist, Research Software Engineering *Mathematics and Computer Science Division, Lemont, IL*

University of Chicago **2023 – Present**
Staff At-Large *Consortium for Advanced Science and Engineering (CASE), Chicago, IL*

Arizona State University **2007 – 2009**
Research and Teaching Assistant *Structural and Computational Mechanics Lab, Tempe, AZ*

Earlier Industry Experience **2005 – 2007**
Project Engineer, Wipro Technologies; Engineering Intern, Tata Motors *India*

Projects

SimBoard, E3SM **2026 – Present**
Contributor *simboard-dev.e3sm.org*

- Web catalog of E3SM simulation runs: case metadata, provenance, and links to each run's diagnostics. FastAPI, PostgreSQL, and React, deployed at NERSC.

UXarray **2022 – Present**
Lead Developer *github.com/UXARRAY/uxarray*

- Lead developer of a Python library for analysis on unstructured climate grids, used at DOE labs, NCAR, and universities.
- Wrote its MCP server, which lets AI agents run UXarray analyses on HPC systems through Globus Compute.
- Made a 300-million-cell, 63 GB mesh loadable in 306 s by deduplicating coordinates before grid construction.
- Fixed 0.68 m of drift in the spherical geometry kernels with error-free arithmetic; 1.67× faster in a Numba hot path by removing hidden heap allocation.

E3SM AI Platform **2026 – Present**
Contributor *github.com/E3SM-Project/e3sm-ai-platform*

- Documentation assistant for E3SM that cites sources or declines to answer. Added Argonne's Argo gateway as a model provider.

ncvis **2026 – Present**
Maintainer *DOE SEATS*

- NetCDF viewer for structured and unstructured grids, in C++. Took over maintenance in 2026.
- github.com/SEATStandards/ncvis.

SPOTTER-AI, DOE Genesis Mission **2026 – Present**
Contributor *PI: B. Nicolae*

- Attention capture and poisoning experiments on cancer drug-response models, for provenance-based integrity checks.

SFNO on Aurora **2025 – 2026**
Software Engineer *ALCF*

- Ported a PyTorch climate emulator to Aurora's Intel GPUs and set up its distributed training baseline.

CANDLE and IMPROVE **2017 – 2025**
Core Contributor *DOE ECP and NCI*

- Built the CANDLE/Supervisor hyperparameter search workflow on Swift/T: 10,000+ training runs on Summit, Theta, and Cori.
- Led the IMPROVE benchmark framework for cancer drug-response models. R&D 100 Award, 2023.

FLASH-X **2019 – 2024**
I/O and Compression Lead *Multiphysics simulation code*

- Asynchronous HDF5 checkpointing with SZ3 and ZFP compression: 40–70% less checkpoint overhead on Summit. R&D 100 Award, 2022.

Urban ECP **2016 – 2019**
Software Lead *With C. Catlett*

- Coupled weather, building energy, and CFD simulations for downtown Chicago.
- Representation and evolution of urban weather boundary conditions in downtown Chicago (*Journal of Building Performance Simulation*, 2020).

MeshKit **2009 – 2016**
Principal Investigator *DOE NEAMS*

- Open-source C++ toolkit for reactor core mesh generation. Its parallel path built a 101-million-element core mesh in about 7 minutes on 712 cores.

Tools

Languages: Python, C++, Fortran, Bash
Scientific Python: NumPy, Xarray, Numba, Dask, Polars, PyTorch
AI systems: MCP servers, agent evaluation, distributed training, hyperparameter search
HPC: MPI, HDF5, NetCDF, SZ3 and ZFP compression, Intel and NVIDIA GPUs, PBS, Slurm
Engineering: CMake, CI/CD, packaging (conda-forge, PyPI), Globus Compute, code review

Education

University of Chicago 2020
M.S. in Computer Science

Arizona State University 2009
M.S. in Structural Engineering
Advisors: Dr. S.D. Rajan (ASU), Dr. Ashok Belegundu (PSU)
Graduate Fellowship Recipient (2007–2009)

Indian Institute of Technology (ISM) Dhanbad 2006
B.Tech. in Mechanical Engineering

Funding

- DOE E3SM (active): SimBoard and the E3SM AI platform.
- DOE SEATS (active): UXarray and ncvis.
- NSF Raijin (active): climate model analysis.
- DOE Genesis Mission, SPOTTER-AI (Phase I, 2026).
- DOE ECP CANDLE (2017 – 2023).
- Urban ECP seed funding (2016 – 2019).
- DOE NEAMS (2009 – 2016): PI, MeshKit.

Awards

- R&D 100 Award, CANDLE (2023).
- R&D 100 Award, FLASH-X (2022).
- ATPESC Scholar, Argonne Training Program on Extreme-Scale Computing (2015).
- Best Paper Award, International Meshing Roundtable (2010).
- University Graduate Fellowship, Arizona State University (2007 – 2009).
- First prize, low-budget car design contest, Minda Ltd. and IIT Kharagpur (2004).

Peer-Reviewed Journal Articles

2026 Chen, H., Ullrich, P. A., Panetta, J., Marsico, D., Hanke, M., **Jain, R.**, Zhang, C., et al. "Accurate and Robust Geometric Algorithms for Regridding on the Sphere." *EGUsphere* (preprint).

2026 Partin, A., Vasanthakumari, P., Narykov, O., Wilke, A., Koussa, N., Jones, S., Zhu, Y., **Jain, R.**, Overbeek, J. C., Fernando, G. D., Sanchez-Villalobos, C., Garcia-Cardona, C., Mohd-Yusof, J., Chia, N., Wozniak, J. M., Ghosh, S., Pal, R., Brettin, T. S., Weil, M. R., and Stevens, R. L. "Benchmarking community drug response prediction models: datasets, models, tools, and metrics for cross-dataset generalization analysis." *Briefings in Bioinformatics* 27(1), bbaf667.

2022 Dubey, A., Weide, K., O'Neal, J., Dhruv, A., Couch, S., Harris, J. A., Klosterman, T., **Jain, R.**, Rudi, J., Messer, B., Pajkos, M., Carlson, J., Chu, R., Wahib, M., Chawdhary, S., Ricker, P. M., Lee, D., Antypas, K., Riley, K. M., Daley, C., Ganapathy, M., Timmes, F. X., Townsley, D. M., Vanella, M., Bachan, J., Rich, P. M., Kumar, S., Endeve, E., Hix, W. R., Mezzacappa, A., and Papatheodore, T. "Flash-X: A multiphysics simulation software instrument." *SoftwareX*.

2022 Harris, J. A., Chu, R., Couch, S. M., Dubey, A., Endeve, E., Georgiadou, A., **Jain, R.**, Kasen, D., Laiu, M. P., Messer, O. E. B., O'Neal, J., Sandoval, M. A., and Weide, K. "Exascale models of stellar explosions: quintessential multi-physics simulation." *International Journal of High Performance Computing Applications*.

2020 **Jain, R.**, Luo, X., Sever, G., Hong, T., and Catlett, C. "Representation and evolution of urban weather boundary conditions in downtown Chicago." *Journal of Building Performance Simulation*.

2018 Wozniak, J. M., **Jain, R.**, Balaprakash, P., Ozik, J., Collier, N. T., Bauer, J., Xia, F., Brettin, T., Stevens, R., Mohd-Yusof, J., Garcia-Cardona, C., Van Essen, B., and Baughman, M. "CANDLE/Supervisor: A workflow framework for machine learning applied to cancer research." *BMC Bioinformatics*.

2014 Mahadevan, V. S., Merzari, E., Tautges, T., **Jain, R.**, Obabko, A., Smith, M., and Fischer, P. "High-resolution coupled physics solvers for analysing fine-scale nuclear reactor design problems." *Philosophical Transactions of the Royal Society A*.

2012 Tautges, T. J. and **Jain, R.** "Creating geometry and mesh models for nuclear reactor core geometries using a lattice hierarchy-based approach." *Engineering with Computers*.

2008 Argod, V., Belegundu, A. D., Aziz, A., Agrawala, V., **Jain, R.**, and Rajan, S. D. "MPI-enabled shape optimization of panels subjected to air blast loading." *International Journal for Simulation and Multidisciplinary Design Optimization*.

Peer-Reviewed Conference and Workshop Proceedings

2026 **Jain, R.** and Jacob, R. "Beyond Tool Execution: Evaluating Scientific MCP Interfaces with UXarray." *1st Workshop on Agentic AI for Large-scale Science (AGENT4SC) at IEEE eScience 2026*, Naples, Italy. Presented September 2026. Artifacts and data: github.com/rajeeja/artifacts_escience_2026_uxarray_mcp.

2026 Wozniak, J. M., **Jain, R.**, and Nicolae, B. "Cancer Data Quality Analysis via Forensic Provenance Capture and Importance Awareness." *Computational Approaches for Cancer Workshop (CAFCW) at SC26*. Submitted.

2025 Gwinn, J., Wozniak, J. M., **Jain, R.**, Zhu, Y., Partin, A., Brettin, T., and Stevens, R. "A Workflow for Error Analysis for Drug Response Prediction via Statistical Standardization and Distribution Analysis." *Workshop on Workflows in Support of Large-Scale Science (WORKS)*.

2024 **Jain, R.**, Tang, H., Dhruv, A., and Byna, S. "Enabling Data Reduction for Flash-X Simulations." *SC24 Workshops: DRBSD-10*.

2024 **Jain, R.**, Wozniak, J. M., Partin, A., Wilke, A., Zhu, Y., Vasanthakumari, P., Narykov, O., Overbeek, J., Weaver, R., Wang, C., Liu, Y., Weil, M. R., Brettin, T., and Stevens, R. "Cross-HPO: Optimizing Neural Networks for Cancer Drug Response Using Hyperparameter Tuning on Multiple Pharmacogenomic Datasets." *Computational Approaches for Cancer Workshop at SC24*.

2023 Wozniak, J. M., **Jain, R.**, Wilke, A., Weaver, R., Partin, A., Brettin, T., and Stevens, R. "An Automation Framework for Comparison of Cancer Response Models Across Configurations." *IEEE e-Science*.

2023 Dhruv, A., **Jain, R.**, O'Neal, J., Weide, K., and Dubey, A. "Framework and Methodology for Verification of a Complex Scientific Simulation Software, Flash-X." *Congress in Computer Science, Computer Engineering, and Applied Computing*.

2022 **Jain, R.**, Tang, H., Dhruv, A., Harris, J. A., and Byna, S. "Accelerating Flash-X Simulations with Asynchronous I/O." *PDSW*.

2021 **Jain, R.**, Shah, A., Mohd-Yusof, J., Wozniak, J. M., Brettin, T., Xia, F., and Stevens, R. "Probing Decision Boundaries in Cancer Data Using Noise Injection and Counterfactual Analysis." *CAFCW at SC21*.

2021 **Jain, R.**, Weide, K., Chawdhary, S., and Klostermann, T. "Checkpoint / Restart for Lagrangian Particle Mesh with AMR in Community Code FLASH-X." *SuperCheck*.

2015 **Jain, R.** and Tautges, T. J. "NEAMS MeshKit: Nuclear Reactor Mesh Generation Solutions." *International Congress on Advances in Nuclear Power Plants*.

2015 **Jain, R.**, Mahadevan, V., and O'Bara, R. "Simplifying Workflow for Reactor Assembly and Full-Core Modeling." *Mathematics and Computation (MC2015)*.

2014 **Jain, R.** and Tautges, T. J. "Generating Unstructured Nuclear Reactor Core Meshes in Parallel." *Procedia Engineering*.

2013 **Jain, R.** and Tautges, T. J. "PostBL: Post-Mesh Boundary Layer Mesh Generation Tool." *International Meshing Roundtable*.

2012 **Jain, R.** and Tautges, T. J. "RGG: Reactor Geometry (and Mesh) Generator." *International Congress on Advances in Nuclear Power Plants*.

2012 Mohanty, S., **Jain, R.**, Majumdar, S., Tautges, T. J., and Srinivasan, M. "Coupled Field Structural Analysis of HTGR Fuel Brick Using Abaqus." *International Congress on Advances in Nuclear Power Plants*.

2010 Tautges, T. J. and **Jain, R.** "Creating Geometry and Mesh Models for Nuclear Reactor Core Geometries Using a Lattice Hierarchy-Based Approach." *19th International Meshing Roundtable*. Best Paper Award.

Conference Abstracts, Tutorials, and Presentations

2027 **Jain, R.** "Accurate Conservative Zonal Averaging for Unstructured Earth-System Grids: A Python Implementation of Robust Spherical Geometry Algorithms." *107th AMS Annual Meeting*. Accepted.

2026 **Jain, R.** and Jacob, R. "Refuse to Guess: A Declared Precondition in a UXarray MCP Server Caught What No Model Checked." *SC26 Research Posters*, St. Louis, MO. Accepted.

2026 **Jain, R.**, Abdulla, D., Jacob, R., Clyne, J., Eroglu, O., Medeiros, B., Chen, H., and Ullrich, P. "A Model Context Protocol Server for Agentic Analysis of Unstructured Earth-System Meshes." Presented at *SciFoundationModels 2026 (SciFM26)*.

2026 Eroglu, O., Chen, H., Chmielowiec, P., Clyne, J., Hannay, C., Jacob, R. L., **Jain, R.**, Medeiros, B., Ullrich, P., and Zarzycki, C. M. "Latest Developments in UXarray: A Python Package Supporting Kilometer-Scale Model Outputs in Native Unstructured Grids." *106th AMS Annual Meeting*.

2026 Clyne, J., Eroglu, O., Jacob, R., **Jain, R.**, Medeiros, B., Ullrich, P., and Zarzycki, C. "UXarray: A Python Package for the Analysis of Kilometer-Scale Atmosphere and Ocean Model Outputs." *EGU General Assembly (EGU26)*.

2025 Eroglu, O., Chen, H., Chmielowiec, P., Hannay, C., Jacob, R. L., **Jain, R.**, Medeiros, B., Ullrich, P., and Zarzycki, C. M. "UXarray: Extending Xarray to Support the Analysis of Native, Kilometer-Scale Unstructured Grids in Python." *105th AMS Annual Meeting*.

2025 Clyne, J., Chen, H., Chmielowiec, P., Eroglu, O., Hannay, C., Jacob, R., **Jain, R.**, Medeiros, B., Ullrich, P., and Zarzycki, C. "UXarray: Extending Xarray for Enhanced Support of Unstructured Grids." *EGU General Assembly*.

2024 **Jain, R.** "Tutorial: UXarray for Analysis of Unstructured Climate Data." *SC24*.

2024 Chmielowiec, P., Chen, H., DeCiampa, C., Eroglu, O., Hannay, C., Jacob, R. L., **Jain, R.**, Loft, R., Medeiros, B., Sun, L., Ullrich, P., and Zarzycki, C. M. "UXarray: Extending Xarray with Support for Unstructured Grids." *104th AMS Annual Meeting*.

2024 Eroglu, O., Chen, H., Chmielowiec, P., Clyne, J., DeCiampa, C., Hannay, C., Jacob, R., **Jain, R.**, Loft, R., Medeiros, B., Sun, L., Ullrich, P., and Zarzycki, C. "UXarray: Extensions to Xarray to Support Unstructured Grids." *EGU General Assembly*.

2024 Clyne, J., Chen, H., Chmielowiec, P., Hannay, C., Jacob, R. L., **Jain, R.**, Medeiros, B., Ullrich, P. A., and Zarzycki, C. M. "UXarray: An Extension to Xarray Supporting Analysis and Visualization of Kilometer-Scale Model Outputs." *AGU Fall Meeting*.

2024 **Jain, R.** "UXarray Tutorial." *ESDS Annual Event*. Video.

2023 **Jain, R.** "UXarray for Unstructured Climate Data." *SciPy 2023*.

2023 **Jain, R.** "Data Reduction for FLASH-X Simulations." *HDF User Group 2023*.

2023 Clyne, J., Chen, H., Chmielowiec, P., DeCiampa, C., Eroglu, O., Hannay, C., Jacob, R. L., **Jain, R.**, Medeiros, B., Sun, L., Ullrich, P. A., and Zarzycki, C. M. "UXarray: A Community-Developed Tool for the Analysis and Visualization of Model Output on Unstructured Grids in Python." *AGU Fall Meeting*.

2023 Jacob, R. L., Ullrich, P. A., Zhang, C., **Jain, R.**, Chen, H., Po-Chedley, S., and Vo, T. "Creating New Standards to Simplify Earth System Model Analysis." *AGU Fall Meeting*.

2019 Sever, G., Obabko, A., Jacob, R., **Jain, R.**, and Catlett, C. "Simulations of Idealized and Real Urban Boundary Layer Environments Using NWP and CFD Models." *18th Conference on Mesoscale Processes*.

2019 Jacob, R. L., Sever, G., Obabko, A., **Jain, R.**, and Catlett, C. E. "CFD Simulations of Flow in Realistic Urban Geometries Initialized from Weather Models." *AGU Fall Meeting*.

2018 Obabko, A., **Jain, R.**, Jacob, R., Paoli, R., Sever, G., Grindeanu, I., and Fischer, P. "Nek5000 Wall-Resolved LES for Urban Geometries." *10th International Conference on Urban Climate / 14th Symposium on the Urban Environment*.

2018 Silva, M. P., Sharma, A., Budhathoki, M., **Jain, R.**, and Catlett, C. E. "Neighborhood Scale Heat Mitigation Strategies Using Array of Things Data in Chicago." *AGU Fall Meeting*.

2018 Sever, G., Jacob, R., and **Jain, R.** "Mesoscale to Microscale Weather Simulations over the Chicago Downtown Region." *10th International Conference on Urban Climate / 14th Symposium on the Urban Environment*.

Technical Reports, Manuals, and Other Publications

2020 Goldring, N., Bruhwiler, D., Nash, B., Wu, Z., Nagler, R., Carter, J., Lerch, J., Suthar, K., Den Hartog, P., **Jain, R.**, and Mahadevan, V. "Multiphysics Design and Optimization of Complex Vacuum Chambers (Phase II Final Technical Report)."

2017 **Jain, R.** "Reactor Geometry (&Mesh) Generator Users Manual."

2016 **Jain, R.**, Vanderzee, E., Grindeanu, I., and Mahadevan, V. "Mesh Generation and Algorithm Development for NEAMS."

2015 **Jain, R.**, Vanderzee, E., and Mahadevan, V. "Update on Development of Mesh Generation Algorithms in MeshKit." ANL/MCS-TM-355.

2015 **Jain, R.** and Mahadevan, V. "Documentation for MeshKit-Reactor Geometry (&mesh) Generator." ANL/MCS-TM-354.

2015 Mahadevan, V., Grindeanu, I. R., Ray, N., **Jain, R.**, and Wu, D. "SIGMA Release v1.2: Capabilities, Enhancements and Fixes." ANL/MCS-TM-357.

2015 Merzari, E., Shemon, E. R., Yu, Y., Thomas, J. W., Obabko, A., **Jain, R.**, Mahadevan, V., Solberg, J., Ferencz, R., and Whitesides, R. "Full Core Multiphysics Simulation with Offline Mesh Deformation." ANL/NE-15/42.

2015 Merzari, E., Shemon, E. R., Yu, Y. Q., Thomas, J. W., Obabko, A., **Jain, R.**, Mahadevan, V., Tautges, T., Solberg, J., Ferencz, R. M., and Whitesides, R. "Multi-physics Demonstration Problem with the SHARP Reactor Simulation Toolkit." ANL/NE-15/44.

2014 **Jain, R.** "Report on FY11 Extensions to MeshKit and RGG." ANL/MCS-TM-316.

2014 **Jain, R.** and Mahadevan, V. "2014 MeshKit Release." ANL/MCS-TM-344.

2014 Tautges, T. J. and **Jain, R.** "Mesh Copy/Move/Merge Tool for Reactor Simulation Applications." ANL/MCS-TM-343.

2013 Tautges, T. J., Fischer, P., Grindeanu, I., **Jain, R.**, Mahadevan, V., Obabko, A., Smith, M. A., Merzari, E., and Ferencz, R. M. "SHARP Assembly-Scale Multiphysics Demonstration Simulations." ANL/NE-13/9.

2013 **Jain, R.** and Tautges, T. J. "MeshKit." ANL/MCS-TM-336.

2012 Tautges, T. J., Fischer, P., Grindeanu, I., **Jain, R.**, Mahadevan, V., Obabko, A., Smith, M. A., Hamilton, S., Clarno, K., and Baird, M. "A Coupled Thermal / Hydraulics – Neutronics – Fuel Performance Analysis of an SFR Fuel Assembly." Report to the U.S. Department of Energy.

Service

- Program committee, AgenticAI4HPC workshop at SC26.
- Program committee, AGENT4SC workshop at IEEE eScience 2026.
- DOE SBIR/STTR proposal reviewer.
- Panelist, "Revolutionizing Public Infrastructure: The Impact of AI and Machine Learning," 5th Infraday Midwest Event.
- Reviewer, *Journal of Open Research Software* and NumGrid.
- Program committee member, NumGrid 2020.
- Session chair, Computational Geometries session, MC2015.
- Volunteer, South Side Science Festival, University of Chicago (2023, 2024).

Memberships

- Association for Computing Machinery (ACM), since 2018.
- American Meteorological Society (AMS).
- American Nuclear Society (ANS), 2011 – 2015.

Mentorship

- Dayan Abdulla (student, 2024 – present): UXarray MCP server.
- Aaron Zedwick (student, 2023 – 2024): UXarray dual mesh and remapping.
- Mark Bartoszek (systems administration, 2023).
- Rylie Weaver (research aide, 2022 – 2024): hyperparameter search for IMPROVE.
- Brett Rhodes (SULI summer student, 2013): MeshKit documentation.